HARSH SAW

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SUMMARY

Software engineer with 2+ years shipping production AI and distributed systems: agentic and RAG platforms, audio AI analysis, Kafka-driven messaging, and Go control planes. Currently building a college-wide agentic AI platform at Okanagan College and audio intelligence at Ommuse.

EXPERIENCE

Software Engineer, AI & Automation | **Okanagan College, IT Services**, Kelowna, BC — Apr 2026 – Present

- Building **OC Connect**, an all-in-one **agentic AI** platform on **Azure AI Foundry** rolling out staff-wide: one place to manage, request, and update leave, finance, payroll, and item requests instead of separate systems.
- Developed its **custom RAG** layer over heterogeneous institutional data, with **microservices** integrating **Banner** SIS and multiple platform APIs behind one retrieval and action interface.
- Shipped the **workflow and issue-tracking system** now used college-wide: Jira-style process tracking with real-time collaboration that replaced manual coordination between staff, instructors, and students.
- Building **ELMS v2**, the next version of an org-wide platform, in parallel with OC Connect.

Software Engineer | **Ommuse**, Seattle, US — Nov 2025 – Present

- Building **AI semantic search** and natural-language chat for a **12K+ user** music platform (18K+ tracks, 120 GB of audio) on a **RAG** pipeline (**MongoDB Vector Store, AWS Bedrock**), letting producers query projects, tracks, and collaborators conversationally.
- Own engineering support for **enterprise customers**, monitoring account usage and in-product activity to catch needs early and act on them.
- Developing a **voice and audio AI analysis agent** that analyzes uploaded tracks and maps the results to structured, queryable track data, extending conversational search from project metadata to the audio itself.
- Helped replace **HubSpot** with **in-house product analytics**, after writing the CRM sync in **Go** (async batch worker, **sync.Map** deduplication).
- Shipped a desktop uploader (**Electron**) with folder sync, parallel **S3 multipart uploads**, **OAuth 2.0 PKCE** auth, and a typed **gRPC** bridge.

Software Engineer | **Bwisher Ltd**, Remote — Dec 2024 – Jun 2025

- Built a multi-channel marketing platform on **Apache Kafka** and **BullMQ** that delivers personalized SMS, email, and **WhatsApp** campaigns with custom offers to **30K+ users** while feeding real-time CRM and analytics pipelines.
- Designed idempotent **NestJS** and **FastAPI** microservices for payment and credit flows, with retry and exponential backoff to eliminate duplicate transactions.
- Owned deployment infrastructure on **Jenkins, Kubernetes, Prometheus,** and **Grafana**; released behind feature flags for tenant-by-tenant rollouts with an instant kill switch.

Full-Stack Developer | **Jythu Ltd**, Remote — Jan 2024 – Dec 2024

- Cut video seek time from **30s to 1.5s** and bandwidth by about **70%** with an **FFmpeg** HLS transcoding pipeline delivered via **S3/CloudFront** and driven by **SQS, Lambda, and SNS**.
- Shipped two production **React Native** apps (LMS + admin panel) with **Node.js** backends serving **2K+ learners**; provisioned the **AWS** infrastructure with **Terraform** and **GitHub Actions** CI/CD, with anti-piracy controls including device-bound auth.

PROJECTS

Remote Orchestrator: Self-Hosted AI Agent Control Plane *Go, TypeScript, Next.js, PostgreSQL, WebSockets, Protobuf, tmux* — GitHub

- Designed a distributed control plane in **Go** that runs Claude Code, Codex, and other CLI agents across self-hosted machines through worker daemons and a versioned **Protocol Buffers** protocol over **WebSockets**.
- Built persistent remote terminal sessions on **PTYs** and **tmux** with live streaming, interactive execution, and reattach after network or control-plane drops.
- Added a fault-tolerant execution lifecycle (heartbeat monitoring, backoff reconnects, leases, cancellation, and resource budgets) with human-in-the-loop approvals and auditable run histories in a **Next.js** dashboard backed by **PostgreSQL**.

Document Intelligence Platform (RAG) *FastAPI, LangChain, Qdrant, Groq, Google Gemini, AWS ECS Fargate, Docker* — Live Demo | GitHub

- Built a multi-tenant RAG platform with session-isolated **Qdrant** indexes for semantic search and conversational Q&A over PDF, DOCX, and TXT; deployed on **AWS ECS Fargate** with idempotent **SHA-256** ingestion fingerprinting to deduplicate embeddings.
- Implemented automatic LLM provider failover (Groq to Gemini) with **LangChain RunnableWithFallbacks**, plus streaming retrieval with context-aware query rewriting and hybrid document chunking.

TECHNICAL SKILLS

AI & ML: LangGraph, LangChain, RAG, agentic workflows, tool calling, LLM evaluation (RAGAS), LangSmith, embeddings, cross-encoder reranking, Qdrant, pgvector, FAISS, Pinecone, AWS Bedrock, Azure AI Foundry, PyTorch, Hugging Face Transformers, PEFT/LoRA/QLoRA, Unsloth, vLLM
Backend & Distributed Systems: NestJS, FastAPI, Express, GraphQL, gRPC/Protobuf, WebSockets, Apache Kafka, BullMQ, RabbitMQ, microservices
Languages: Go (Golang), Python, TypeScript, JavaScript (ES6+), Bash
Databases: PostgreSQL, MongoDB, DynamoDB, Redis, MySQL, SurrealDB, SQLite
Cloud & DevOps: AWS (ECS Fargate, Lambda, SQS, S3), Azure, Docker, Kubernetes, Terraform, Pulumi, Jenkins, GitHub Actions, Prometheus, Grafana
Frontend & Mobile: React, Next.js, React Native, Qwik.js, Electron

ACHIEVEMENTS

- Built an end-to-end stock prediction pipeline covering **500 S&P 500 constituents** over 5 years of 15-minute bar data, with **XGBoost** models trained across 26 intraday horizons on **4x NVIDIA H100 GPUs** (DRAC), automated warm-start retraining every 15 minutes during market hours, and a **FastAPI** backend backed by a **PostgreSQL** data warehouse and a multi-rolling feature pipeline.
- Sole developer of an internal LMS migration platform at **Okanagan College**, built and deployed end-to-end and relied on daily by **1,500+ staff and instructors** for the institution-wide Moodle-to-Brightspace transition, featuring guided step-by-step migration workflows with progress tracking and complete records, real-time notifications and chat, and AI-powered templating.
- Ranked in the **top 2% of verified freelancers** on Freelancer.com with a 5-star rating, delivering 25+ production-grade projects.

EDUCATION

Bachelor of Computer Information Systems, Okanagan College — 2022 - 2026